

Question number	Answer	Additional guidance	Mark
2(a)(i)	D is the correct answer A is not the correct answer the sodium channel is not closed in depolarisation B is not the correct answer the sodium channel is not closed in depolarisation C is not the correct answer as the membrane potential is not decreasing in depolarisation		(1)

Question number	Answer	Additional guidance	Mark
2(a)(ii)	An explanation that includes the following points: - sodium (ion) {channels / voltage-gated channels} close (1) - potassium ions move out of cell via (voltage gated) potassium channels (1) - the inside of the axon becomes negatively charged (compared with the outside) (1)	accept (voltage-gated) potassium (ion) channels open reject contradictions e.g. potassium ion moving in through open channels {decrease in membrane potential / back to (below)-70mv} ignore back to resting potential	(3)

Question number	Answer	Additional guidance	Mark
2(b)(i)	A calculation showing the following steps: • measurement of scale bar converted to μm (1) • calculation of magnification and answer given in standard form (1)	9.5mm = 9500 (μm) ACCEPT 9000 to 10000 (μm) $(9500 \div 50) = 190 = 1.9 \times 10^2$ accept 1.8×10^2 to 2.0×10^2 accept 2×10^2 180 to 200 gets 1 mark correct answer gains full marks	(2)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	An answer that includes: • light (microscope) (1)	mark first answer	(1)

Question number	Answer	Additional guidance	Mark
2(b)(iii)	An explanation that includes three of the following points: - ions cannot cross axon membrane in myelinated regions (1) - as myelin acts as an insulator (1) - therefore {action potentials / depolarisation} (only) occur at nodes of Ranvier (1) - so {nerve impulses / action potentials} jump from node to node / saltatory conduction (1)	ignore myelinated neurone {has / creates} insulation accept gaps between local currents are longer	(3)